

Four-Channel Compression in a Primitive Möbius Tail: Centered Divisor Kernels, a Single-Leg Completion Wedge, and CRT-Fiber Discrepancy

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Abstract

TPC-19 shows that the primitive residual correlation left by TPC-18 cannot be estimated by centering the prime–prime, prime–model, model–prime, and model–model channels separately. We give the missing common algebraic reduction and then sharpen its long-period spectral gate.

Let P_m and M_m be the locally calibrated prime and finite Ramanujan-model signals on a primitive affine row $mj + h$, put $H = \text{rad } |h|$, and set $Z_m = P_m - M_m$. For a finite target horizon U , we prove the pointwise identity

$$Z_m(j) = \varepsilon_U(m) + \sum_{\substack{u \leq U \\ (u, mH)=1}} b_R(u) \left(\mathbf{1}_{u|mj+h} - \frac{1}{u} \right), \quad b_R(u) = -\mu(u)w_R(u),$$

where $w_R(u) > 0$ is explicit. A uniform Möbius calibration theorem gives $\varepsilon_U(m) \ll_A \rho_{P,H}(m)(\log U)^{-A}$ throughout every fixed polynomial row range. Thus the full $PP - PM - MP + MM$ product has one coefficient and one centered divisor kernel, up to an arbitrarily logarithmically small scalar defect. Its primitive-conditional centered-kernel (u, v) -zero mode is

$$\frac{b_R(u)b_R(v)}{uv} ((u, v) \mathbf{1}_{(u,v)|m_1-m_2} - 1),$$

so the zero mode of every coprime divisor pair vanishes termwise in the matched channel.

For $g = (u, v)$, $u = ga$, $v = gb$, and $l = [u, v] = gab$, we compute every Fourier coefficient of the primitive centered kernel on its true period Hl . When the two divisor congruences are compatible, all four phases align and the coefficient contains the exact multiplier

$$B_{g,a,b}(r) = g - \mathbf{1}_{b|r} - \mathbf{1}_{a|r} + \mathbf{1}_{l|r}.$$

This yields a two-scale Poisson theorem. If $H \max(u, v) \leq JX^{-\eta}$, the two single-indicator spectra and the constant spectrum complete with arbitrary power accuracy even when the joint period Hl is long. For all $u, v \leq R = X^{1/2-\delta}$, the strict-margin range in which every single and constant spectrum completes is $Q \leq X^{1/2+\delta-\eta_0}$, substantially wider than the $Q \leq X^{2\delta-\eta_0}$ range obtained by completing the whole joint period at once. In the genuinely remaining low-frequency joint packet, however, $B_{g,a,b}(r) = g$; centering supplies no further nonzero-frequency cancellation.

Finally, compatible row pairs are aggregated by their joint CRT class $\kappa \bmod l$. The three inverse phases of TPC-19 compress exactly to the ordinary finite Fourier transform of a single

fiber vector $Z_l(\kappa)$. Splitting that vector into its unit-class mean and discrepancy produces an explicit Ramanujan bias and an exact Parseval identity for the fiber variance. For this chosen compatible short-short packet, the remaining inputs are a bound for the explicit Ramanujan mean and a quantitative estimate for this modulus-dependent fiber discrepancy, or a norm-controlled low-rank substitute. The cited large-sieve and Kloosterman-fraction theorems do not automatically provide those inputs for the sharp Möbius row weights, and the large-divisor legs remain outside this reduction. No residual-dispersion estimate, fixed-shift Hardy–Littlewood asymptotic, twin-prime theorem, or breach of sieve parity is claimed.

Keywords: Möbius tail; centered divisor kernel; determinant spectrum; Chinese remainder theorem; Poisson summation; CRT fibers; Ramanujan sums; sieve parity.

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1 Introduction

TPC-18 reduces a failed symmetric fixed-shift tail block to a primitive, generic two-row correlation of the residual $r_R = \Lambda - \Lambda_R$, where

$$\Lambda_R(n) = \sum_{q \leq R} \frac{\mu(q)}{\varphi(q)} c_q(n)$$

and the row determinant is $h(m_1 - m_2)$ [9]. TPC-19 then gives exact divisor and Poisson normal forms for this correlation [10]. It also records a crucial negative fact: the centered model-model channel can have a nonzero complete-period mean on a primitive, distinct-source row pair. A proof cannot therefore estimate

$$PP, \quad PM, \quad MP, \quad MM$$

independently after taking absolute values. Either additional oscillation over the whole row family or a genuinely matched spectral argument is required.

The first purpose of this paper is to carry out that matching at the algebraic level. Let P_m and M_m denote the locally calibrated prime and finite-model signals on the row $mj + h$, and put $Z_m = P_m - M_m$. The four products above are exactly the expansion of $Z_{m_1} Z_{m_2}$. We prove that, on every finite target horizon, this difference has one common centered-divisor representation

$$Z_m(j) = \varepsilon_U(m) + \sum_u b_R(u) C_{m,u}(j), \quad C_{m,u}(j) = \mathbf{1}_{u|mj+h} - \frac{1}{u}. \quad (1)$$

Here $b_R(u) = -\mu(u)w_R(u)$ is the actual residual coefficient found in TPC-19, not the coefficient of only one of the four channels. The scalar $\varepsilon_U(m)$ is not discarded formally: we prove a uniform Möbius calibration theorem which makes it arbitrarily logarithmically small for every polynomial row range. Consequently the (u, v) -term of the matched centered kernel has the primitive-conditional connected zero mode

$$\frac{b_R(u)b_R(v)}{uv} \left((u, v) \mathbf{1}_{(u,v)|m_1-m_2} - 1 \right). \quad (2)$$

The zero mode of every $(u, v) = 1$ term vanishes before any estimate is made.

The second purpose is to resolve the apparent three-phase complexity of the long-lcm Poisson spectrum. Write

$$g = (u, v), \quad u = ga, \quad v = gb, \quad q = [u, v] = gab.$$

For compatible divisor congruences, all four centered Fourier terms align on the common period Hq , where $H = \text{rad } |h|$. Their exact multiplier is

$$B_{g,a,b}(r) = g - \mathbf{1}_{b|r} - \mathbf{1}_{a|r} + \mathbf{1}_{q|r}. \quad (3)$$

This alignment has two consequences. First, the joint, two single, and constant terms should not all be completed at the longest period. The single terms have periods Hu and Hv . Completing those shorter periods yields a strict-margin range

$$Q \leq X^{1/2+\delta-\eta_0} \quad (4)$$

for completing all single and constant terms across $u, v \leq R = X^{1/2-\delta}$, compared with $Q \leq X^{2\delta-\eta_0}$ when the full joint period is completed at once. The compatible joint packet is retained. Second, on its genuinely remaining low nonzero frequencies, the multiplier in (3) is exactly g . The matched algebra has removed the spurious single spectra, but it has not created cancellation in the surviving joint packet.

The third purpose is to identify that packet without inverse-phase notation obscuring its coefficient. Compatible row residue pairs are parametrized by one unit class $\kappa \bmod q$. After aggregating the actual row weights on these fibers, the joint phase is the ordinary DFT

$$S_q(r) = \sum_{\kappa \in (\mathbb{Z}/q\mathbb{Z})^\times} Z_q(\kappa) e_q(r\overline{H}\kappa). \quad (5)$$

The three inverse phases displayed in TPC-19 are precisely the CRT factorization of this one character. Splitting Z_q into its unit-class mean and discrepancy gives

$$S_q(r) = \overline{Z}_q c_q(r) + \widehat{D}_q(r)$$

and the exact identity

$$\sum_{r \bmod q} |\widehat{D}_q(r)|^2 = q \sum_{\kappa \in (\mathbb{Z}/q\mathbb{Z})^\times} |D_q(\kappa)|^2. \quad (6)$$

Thus, for this chosen compatible short-short packet, the remaining gates are a quantitative theorem for the actual prime-Möbius row weights across CRT fibers and a separate bound for the explicit Ramanujan mean packet. The large-divisor legs $u > R$ or $v > R$ remain outside this reduction. The fiber gate is not an unspecified need for cancellation among three unrelated inverse phases.

These reductions are rigorous progress, but they do not supply the required estimates for this packet. A coefficient concentrated on one CRT fiber saturates Parseval. At the opposite extreme, a uniform fiber vector produces an explicit Ramanujan bias, and on coprime frequencies its transverse Möbius sign migrates to the determinant content. The cited bilinear and trilinear Kloosterman-fraction estimates apply only after a separable coefficient reduction, while the current fiber vector is a modulus-dependent joint array [1, 3, 11]. Pascadi's frequency-concentration theorem gives a powerful abstract interface once a representing measure satisfying its size and two-frequency conditions has been constructed; it does not prove those conditions for the sharp coefficient here [7].

The organization is as follows. [Section 2](#) fixes the primitive row geometry. [Section 3](#) proves the common four-channel coefficient, and [section 4](#) controls its scalar calibration defect uniformly and proves packet-scale negligibility under explicit packet hypotheses, which are satisfied for the separated packets in the single-leg range. [Section 5](#) computes the complete Fourier kernel. [Section 6](#) proves the new completion wedge and its sharp smoothness boundary. [Section 7](#) gives the fiber DFT, Ramanujan mean, variance identity, and no-go results. Current technology is audited in [section 8](#); the exact certificate and conclusion are in [section 9](#).

2 Primitive rows and locally calibrated signals

Fix $h \in \mathbb{Z} \setminus \{0\}$ and put

$$H = \text{rad } |h|. \quad (7)$$

Let $0 < \delta < 1/2$ and

$$R = \lfloor X^{1/2-\delta} \rfloor, \quad V = \lfloor R^{1/2} \rfloor. \quad (8)$$

Throughout X is sufficiently large in terms of h and δ that $R \geq H^2$. Hence $dH \leq R$ for every $d \leq V$ in the row support below. The finite Ramanujan model and its residual are

$$\Lambda_R(n) = \sum_{q \leq R} \frac{\mu(q)}{\varphi(q)} c_q(n), \quad r_R(n) = \Lambda(n) - \Lambda_R(n). \quad (9)$$

We write $e_q(x) = \exp(2\pi i x/q)$, set $e_1(x) = 1$, and use $c_q(n) = \sum_{a \bmod q}^* e_q(an)$.

Let L, D, J, Q satisfy

$$Q = LD, \quad QJ \asymp X, \quad L > \max(2R, 2|h|), \quad (10)$$

and let

$$\ell \in [L/2, 2L] \text{ be prime}, \quad d \in [D, 2D] \cap [1, V] \quad (11)$$

with d squarefree. In the primitive sector

$$(d, H) = 1, \quad (j, H) = 1. \quad (12)$$

Put

$$m = \ell d, \quad n_m(j) = mj + h. \quad (13)$$

Then $(m, H) = 1$, and every divisor $u \mid n_m(j)$ satisfies

$$(u, mH) = 1. \quad (14)$$

TPC-18 restricts the final primitive obstruction to generic row pairs $m_i = \ell_i d_i$ satisfying

$$\ell_1 \neq \ell_2, \quad |m_1 - m_2| > QX^{-\kappa}, \quad (d_1, d_2) \leq X^\kappa \quad (15)$$

for a fixed $\kappa > 0$. All exact identities below hold before this pruning; any row-pair restriction independent of j may therefore be imposed outside the identities. Pruning does not, however, preserve a Gram or positive-semidefinite structure.

The exact primitive conditional mean of the finite model is

$$\rho_{M,H,R}(m) = \frac{H}{\varphi(H)} \frac{d}{\varphi(d)}, \quad (16)$$

while the calibrated prime local density is

$$\rho_{P,H}(m) = \frac{H}{\varphi(H)} \frac{m}{\varphi(m)} = \frac{mH}{\varphi(mH)}. \quad (17)$$

The second formula is a local calibration, not a prime number theorem for the growing modulus mH . Define

$$P_m(j) = \Lambda(mj + h) - \rho_{P,H}(m), \quad M_m(j) = \Lambda_R(mj + h) - \rho_{M,H,R}(m), \quad (18)$$

and

$$Z_m(j) = P_m(j) - M_m(j). \quad (19)$$

TPC-19 proves the exact drift identity

$$r_R(mj + h) = Z_m(j) + \delta_H(m), \quad \delta_H(\ell d) = \frac{H}{\varphi(H)} \frac{d}{\varphi(d)(\ell - 1)}. \quad (20)$$

The contribution of one drift to a full primitive row-pair correlation is $O_{h,\varepsilon}(XDX^\varepsilon)$, and the two-drift contribution is $O_{h,\varepsilon}(XDL^{-1}X^\varepsilon)$. Both have a fixed power saving relative to the natural scale

$$Q^2 J \asymp XQ \quad (21)$$

whenever $L \geq X^{\lambda_0}$ for fixed $\lambda_0 > 0$.

Throughout, F belongs to a uniformly bounded subset of $C_c^\infty(\mathbb{R})$, and its support is chosen so that the target values are positive and $n_m(j) \ll_F X$. The actual TPC-18 weights also depend smoothly on mj/X and dj/K . Fourier inversion in logarithmic coordinates writes them as an absolutely convergent superposition of separated row and orbit weights; truncating the dual variables at a fixed power of X has arbitrary power accuracy. It therefore suffices to prove the spectral identities for one separated packet with a common factor $F(j/J)$, then restore that superposition.

3 Exact recombination of the four centered channels

We begin with the divisor coefficient of the finite model. Put

$$\lambda'_R(u) = u \sum_{b \leq R/u} \frac{\mu(ub)\mu(b)}{\varphi(ub)} \quad (u \leq R), \quad (22)$$

and set $\lambda'_R(u) = 0$ for $u > R$. The divisor formula for Ramanujan sums gives

$$\Lambda_R(n) = \sum_{u|n} \lambda'_R(u). \quad (23)$$

The coefficient is squarefree supported. For squarefree $u \leq R$,

$$\lambda'_R(u) = \mu(u) \frac{u}{\varphi(u)} \sum_{\substack{b \leq R/u \\ (b,u)=1}} \frac{\mu(b)^2}{\varphi(b)}. \quad (24)$$

Define

$$a(u) = -\mu(u) \log u, \quad b_R(u) = a(u) - \lambda'_R(u). \quad (25)$$

For squarefree u , put

$$w_R(u) = \log u + \mathbf{1}_{u \leq R} \frac{u}{\varphi(u)} \sum_{\substack{b \leq R/u \\ (b,u)=1}} \frac{\mu(b)^2}{\varphi(b)}. \quad (26)$$

Then $w_R(u) > 0$ and

$$b_R(u) = -\mu(u)w_R(u), \quad (27)$$

with both sides understood as zero off the squarefree integers.

Let $U \geq R$ be an integer such that every target in the separated packet under consideration satisfies

$$1 \leq n_m(j) = mj + h \leq U. \quad (28)$$

For $(u, mH) = 1$, define

$$C_{m,u}(j) = \mathbf{1}_{u|mj+h} - \frac{1}{u}, \quad (29)$$

and define the finite prime-calibration defect

$$\varepsilon_U(m) = \sum_{\substack{u \leq U \\ (u, mH)=1}} \frac{a(u)}{u} - \rho_{P,H}(m). \quad (30)$$

Theorem 3.1 (One coefficient for all four channels). *Under the setup of [section 2](#), in particular $dH \leq R$, and under [equation \(28\)](#), one has pointwise*

$$\boxed{Z_m(j) = \varepsilon_U(m) + \sum_{\substack{u \leq U \\ (u, mH)=1}} b_R(u) C_{m,u}(j).} \quad (31)$$

Consequently, with $\varepsilon_i = \varepsilon_U(m_i)$,

$$\begin{aligned} Z_{m_1}(j) Z_{m_2}(j) &= \varepsilon_1 \varepsilon_2 + \varepsilon_1 \sum_v b_R(v) C_{m_2,v}(j) + \varepsilon_2 \sum_u b_R(u) C_{m_1,u}(j) \\ &\quad + \sum_{u,v} b_R(u) b_R(v) C_{m_1,u}(j) C_{m_2,v}(j), \end{aligned} \quad (32)$$

where the last three sums carry the horizon and coprimality restrictions from [\(31\)](#). Thus the product

$$P_{m_1} P_{m_2} - P_{m_1} M_{m_2} - M_{m_1} P_{m_2} + M_{m_1} M_{m_2}$$

has a common coefficient and a common centered kernel before any absolute value is taken.

Proof. The von Mangoldt divisor identity and [\(28\)](#) give

$$\Lambda(n_m(j)) = \sum_{\substack{u \leq U \\ u|n_m(j)}} a(u).$$

Every divisor in this sum is coprime to mH by [\(14\)](#). Adding and subtracting the density of each divisibility condition gives

$$P_m(j) = \varepsilon_U(m) + \sum_{\substack{u \leq U \\ (u, mH)=1}} a(u) C_{m,u}(j). \quad (33)$$

Similarly, [\(23\)](#) and the exact conditional mean [\(16\)](#) give

$$M_m(j) = \sum_{\substack{u \leq R \\ (u, mH)=1}} \lambda'_R(u) C_{m,u}(j). \quad (34)$$

Indeed, averaging the divisor indicators over the primitive classes shows that the sum of $\lambda'_R(u)/u$ in [\(34\)](#) is exactly $\rho_{M,H,R}(m)$. Subtracting [\(34\)](#) from [\(33\)](#) proves [\(31\)](#); multiplication proves [\(32\)](#). $\square$

Remark 3.2 (What has and has not been matched). [Theorem 3.1](#) is a finite, pointwise identity. It constructs the common divisor coefficient which TPC-19 left as an analytic requirement. It is not a finite-window prime number theorem in progressions and does not estimate the resulting two-row sum. The only scalar not absorbed into the common kernel is $\varepsilon_U(m)$; it is controlled uniformly in [section 4](#).

For a periodic function of j , let $\mathbb{E}_H$ denote the complete-period average conditional on $(j, H) = 1$.

Proposition 3.3 (Connected determinant zero mode). *Let $(u, m_1 H) = (v, m_2 H) = 1$, put*

$$g = (u, v), \quad \Delta = m_1 - m_2.$$

Then

$$\mathbb{E}_H[C_{m_1, u} C_{m_2, v}] = \frac{1}{uv} (g \mathbf{1}_{g|\Delta} - 1). \quad (35)$$

In particular, every $g = 1$ zero mode in the recombined $PP - PM - MP + MM$ channel vanishes termwise. Moreover,

$$g \mathbf{1}_{g|\Delta} - 1 = \sum_{r=1}^{g-1} e_g(r\Delta). \quad (36)$$

Proof. The two congruences for j are individually soluble and have conditional densities $1/u$ and $1/v$. Their joint density is $g/(uv)$ precisely when their reductions modulo g agree. Primitivity makes this compatibility equivalent to $g \mid \Delta$; otherwise the joint density is zero. Expanding the two centered indicators proves (35). The complete additive-character sum modulo g , with its zero frequency removed, gives (36). $\square$

The common coefficient also identifies exactly where the Möbius signs remain. On squarefree support write

$$u = ga, \quad v = gb, \quad g, a, b \text{ pairwise coprime.} \quad (37)$$

Then

$$b_R(ga) b_R(gb) = \mu(a) \mu(b) w_R(ga) w_R(gb). \quad (38)$$

The determinant content g is unsigned. If the TPC-18 row signs $\mu(d_1) \mu(d_2)$ are also included, the admissibility conditions imply $(a, d_1) = (b, d_2) = 1$, and hence

$$\mu(d_1) \mu(d_2) \mu(a) \mu(b) = \mu(d_1 a) \mu(d_2 b). \quad (39)$$

No coprimality between $d_1 a$ and $d_2 b$ is asserted: d_1, d_2 may intersect, as may a, d_2 and b, d_1 .

4 Uniform removal of the finite calibration defect

The formal identity

$$\sum_{\substack{n \geq 1 \\ (n, M)=1}} \frac{-\mu(n) \log n}{n} = \frac{M}{\varphi(M)}$$

is a conditionally convergent statement at $s = 1$. Since the row modulus grows with X , a fixed- M limit is not enough for (30). We prove the required uniform form.

Theorem 4.1 (Uniform Möbius calibration). *For every fixed $A > 0$ and $C > 0$, uniformly for*

$$U \geq 3, \quad 1 \leq M \leq U^C, \quad (40)$$

one has

$$\sum_{\substack{n \leq U \\ (n, M)=1}} \frac{-\mu(n) \log n}{n} = \frac{M}{\varphi(M)} + O_{A, C} \left(\frac{M}{\varphi(M)} (\log U)^{-A} \right). \quad (41)$$

Squarefreeness of M is not required.

Proof. Let

$$\mathcal{S}(M) = \{k \geq 1 : p \mid k \implies p \mid M\}, \quad g_M(k) = \mathbf{1}_{k \in \mathcal{S}(M)}.$$

Its Dirichlet series is

$$G_M(s) = \sum_{k \in \mathcal{S}(M)} k^{-s} = \prod_{p \mid M} (1 - p^{-s})^{-1}, \quad G_M(1) = \frac{M}{\varphi(M)}. \quad (42)$$

Euler factors give the exact convolution

$$\mu(n) \mathbf{1}_{(n, M)=1} = (\mu * g_M)(n). \quad (43)$$

Put

$$\mathcal{A}(x) = \sum_{n \leq x} \frac{-\mu(n) \log n}{n}, \quad \mathcal{B}(x) = \sum_{n \leq x} \frac{\mu(n)}{n}.$$

Opening (43) and using $-\log(dk) = -\log d - \log k$ gives

$$\begin{aligned} S_M(U) &:= \sum_{\substack{n \leq U \\ (n, M)=1}} \frac{-\mu(n) \log n}{n} \\ &= \sum_{\substack{k \leq U \\ k \in \mathcal{S}(M)}} \frac{\mathcal{A}(U/k)}{k} - \sum_{\substack{k \leq U \\ k \in \mathcal{S}(M)}} \frac{\log k}{k} \mathcal{B}(U/k). \end{aligned} \quad (44)$$

Therefore

$$\begin{aligned} S_M(U) - G_M(1) &= \sum_{\substack{k \leq U \\ k \in \mathcal{S}(M)}} \frac{\mathcal{A}(U/k) - 1}{k} \\ &\quad - \sum_{\substack{k \leq U \\ k \in \mathcal{S}(M)}} \frac{\log k}{k} \mathcal{B}(U/k) - \sum_{\substack{k > U \\ k \in \mathcal{S}(M)}} \frac{1}{k}. \end{aligned} \quad (45)$$

The classical zero-free region for $\zeta(s)$, followed by partial summation, implies that for every $B > 0$

$$|\mathcal{A}(x) - 1| + |\mathcal{B}(x)| \ll_B (\log x)^{-B} \quad (x \geq 3). \quad (46)$$

See, for example, Iwaniec and Kowalski [5, Chapter 5]. The two limiting constants follow respectively from $(1/\zeta)'(1) = 1$ and $1/\zeta(1) = 0$. For $x \geq 1$, the crude bounds

$$|\mathcal{A}(x) - 1| \ll 1 + \log^2(2x), \quad |\mathcal{B}(x)| \ll \log(2x) \quad (47)$$

will suffice.

Write $\mathcal{L} = \log U$ and choose

$$\eta = \frac{C_0 \log \mathcal{L}}{\mathcal{L}}, \quad P_0 = \exp\left(\frac{\mathcal{L}}{4C_0 \log \mathcal{L}}\right), \quad (48)$$

where C_0 will be fixed at the end. We claim

$$G_M(1 - \eta) \ll_{C, C_0} G_M(1). \quad (49)$$

Indeed,

$$\log \frac{G_M(1 - \eta)}{G_M(1)} = \int_{1-\eta}^1 \sum_{p \mid M} \frac{\log p}{p^\sigma - 1} d\sigma.$$

For $p \leq P_0$, the definition of P_0 gives $\eta \log p \leq 1/4$, and the contribution is

$$\ll \eta \sum_{p \leq P_0} \frac{\log p}{p} \ll 1.$$

There are at most $\log M / \log P_0 \ll_{C, C_0} \log \mathcal{L}$ prime divisors $p > P_0$, and their total contribution is

$$\ll_{C, C_0} (\log \mathcal{L}) P_0^{-1+\eta} = o(1).$$

This proves (49). The same split gives

$$\mathcal{L}_M(1 - \eta) := \sum_{p|M} \frac{\log p}{p^{1-\eta} - 1} \ll_{C, C_0} \mathcal{L}. \quad (50)$$

Rankin's trick, (49), and (50) now give, for $Y \geq 1$,

$$\sum_{\substack{k > Y \\ k \in \mathcal{S}(M)}} \frac{1}{k} \ll Y^{-\eta} G_M(1), \quad (51)$$

$$\sum_{\substack{k > Y \\ k \in \mathcal{S}(M)}} \frac{\log k}{k} \ll Y^{-\eta} G_M(1) \mathcal{L}. \quad (52)$$

Set

$$Z = \exp(\sqrt{\mathcal{L}}), \quad Y_0 = U/Z.$$

For $k \leq Y_0$, one has $\log(U/k) \geq \sqrt{\mathcal{L}}$. Applying (46) with exponent B_0 , and using

$$\sum_{k \in \mathcal{S}(M)} \frac{\log k}{k} = -G'_M(1) \ll_C G_M(1) \mathcal{L},$$

bounds the two central pieces of (45) by

$$O\left(G_M(1) \mathcal{L}^{-B_0/2}\right) + O\left(G_M(1) \mathcal{L}^{1-B_0/2}\right). \quad (53)$$

Moreover,

$$\eta \log Y_0 \geq \frac{C_0}{2} \log \mathcal{L}$$

for sufficiently large U . Equations (51) and (52), combined with the crude bounds (47) on $Y_0 < k \leq U$, bound all three tail pieces by

$$O\left(G_M(1) \mathcal{L}^{3/2-C_0/2}\right). \quad (54)$$

Taking $B_0 = C_0 = 2A + 8$, and enlarging the constant for bounded U , proves (41). $\square$

Corollary 4.2 (The scalar defect in a separated TPC packet). *Take $U \asymp_F X$ large enough for (28). Uniformly over the primitive row family,*

$$\varepsilon_U(m) \ll_{A, h, F} \rho_{P, H}(m) (\log X)^{-A} \ll_{A, h, F, \epsilon} X^\epsilon (\log X)^{-A}. \quad (55)$$

Let

$$\tilde{Z}_m(j) = \sum_{\substack{u \leq U \\ (u, mH)=1}} b_R(u) C_{m, u}(j). \quad (56)$$

Let $\mathcal{M}$ be any set of rows in (11), so that $\#\mathcal{M} \ll Q$, and define

$$\mathcal{C}_\omega(A, B) = \sum_{m_1, m_2 \in \mathcal{M}} \omega_{m_1, m_2} \sum_{\substack{j \in \mathbb{Z} \\ (j, H)=1}} F(j/J) A_{m_1}(j) B_{m_2}(j).$$

For a fixed $K \geq 0$, assume

$$\max_{m_1, m_2} |\omega_{m_1, m_2}| \leq (\log X)^K, \quad R \leq J(\log X)^K. \quad (57)$$

Then replacing Z by $\tilde{Z}$ changes the packet by

$$\mathcal{C}_\omega(Z, Z) - \mathcal{C}_\omega(\tilde{Z}, \tilde{Z}) = O_{B, K, h, F}(XQ(\log X)^{-B}) \quad (58)$$

for every prescribed $B > 0$. In particular, the second condition in (57) holds throughout the whole short-short strict-margin wedge of corollary 6.4.

Proof. Apply theorem 4.1 with $M = mH$. The row range is polynomial in X , so a fixed C covers every row. Also $M/\varphi(M) = \rho_{P, H}(m) \ll_\epsilon X^\epsilon$; the sharper standard bound $\rho_{P, H}(m) \ll_h \log \log(3X)$ will be used below. By theorem 3.1, $Z_m - \tilde{Z}_m = \varepsilon_U(m)$.

We first record an elementary L^1 bound with no prime-distribution input. From (24),

$$\sum_{u \leq R} \frac{|\lambda'_R(u)|}{u} \leq \sum_{\substack{ub \leq R \\ (u, b)=1}} \frac{\mu(u)^2 \mu(b)^2}{\varphi(u) \varphi(b)} \ll (\log(2R))^2, \quad (59)$$

$$\sum_{u \leq R} |\lambda'_R(u)| \ll R(\log(2R))^2. \quad (60)$$

The first estimate follows by majorizing with the Euler product $\prod_{p \leq R} (1 + 2/(p-1)) \ll (\log(2R))^2$; the second follows by multiplying each summand by $u \leq R$. Every u with a nonempty divisor congruence is coprime to m by (14); that congruence has at most $O_F(J/u + 1)$ solutions on the support of $F(j/J)$. Therefore

$$\sum_j |F(j/J) \Lambda_R(mj + h)| \ll_F (J + R)(\log X)^2.$$

The trivial inequality $\Lambda(n) \leq \log n$, the $O_F(J)$ orbit support, and the standard bounds for the two local densities now give

$$\sum_j |F(j/J) Z_m(j)| + \sum_j |F(j/J) \tilde{Z}_m(j)| \ll_{h, F} (J + R)(\log X)^2. \quad (61)$$

Expanding

$$Z_{m_1} Z_{m_2} - \tilde{Z}_{m_1} \tilde{Z}_{m_2} = \varepsilon_U(m_1) \tilde{Z}_{m_2} + \varepsilon_U(m_2) \tilde{Z}_{m_1} + \varepsilon_U(m_1) \varepsilon_U(m_2)$$

and applying (61), $\#\mathcal{M} \ll Q$, and (57) bounds the absolute packet error by

$$\ll_{A, K, h, F} Q^2 (\log X)^K \left\{ (\log \log(3X)) (\log X)^{-A} (J + R) (\log X)^2 + J (\log \log(3X))^2 (\log X)^{-2A} \right\}.$$

Use $R \leq J(\log X)^K$, choose A in terms of B, K , and invoke $JQ^2 \asymp XQ$ to prove (58). $\square$

Remark 4.3 (No growing-modulus prime theorem is used). Theorem 4.1 concerns only the ordinary Möbius Dirichlet series and finite Euler factors at primes dividing mH . It is not a prime number theorem in a progression of modulus mH . The genuinely difficult finite-window prime information remains inside the common centered kernel after $\varepsilon_U(m)$ has been removed.

5 The exact four-channel Fourier kernel

We first keep all four terms created by centering. The calculation is coefficient-independent: it applies both to the finite-model coefficient and to every finite short-short block of the common residual coefficient in [theorem 3.1](#). It does not yet estimate the sum over rows or the complementary large-divisor legs.

Put

$$g = (u, v), \quad a = \frac{u}{g}, \quad b = \frac{v}{g}, \quad q = [u, v] = gab, \quad (62)$$

so that $(a, b) = 1$. The finite-model coefficients are squarefree supported, in which case g, a, b are pairwise coprime, but the first Fourier identities below do not require squarefreeness. We assume

$$(u, m_1 H) = (v, m_2 H) = 1. \quad (63)$$

Since $H = \text{rad } |h|$, this also makes h invertible modulo u and v . Define

$$\kappa_1 \equiv -h\overline{m_1} \pmod{u}, \quad \kappa_2 \equiv -h\overline{m_2} \pmod{v}. \quad (64)$$

Every inverse below is taken in the modulus of the character in which it occurs. We use $e_s(x) = \exp(2\pi i x/s)$, with $e_1(x) = 1$, and write

$$c_H(r) = \sum_{c \bmod H}^* e_H(rc). \quad (65)$$

Lemma 5.1 (Primitive compatibility). *The classes in (64) are compatible modulo g if and only if*

$$g \mid m_1 - m_2. \quad (66)$$

When this holds, they determine one class $\kappa \bmod q$, and $(\kappa, q) = 1$.

Proof. Modulo g , compatibility is equivalent to $h(\overline{m_1} - \overline{m_2}) \equiv 0$. Since h, m_1, m_2 are units modulo g , this is equivalent to $m_2 - m_1 \equiv 0 \pmod{g}$. The generalized Chinese remainder theorem gives the class modulo $[u, v] = q$. Its reductions are a unit times h , so it is a unit modulo both u and v , hence modulo q . $\square$

For the two centered congruence indicators set

$$C_{m,u}(j) = \mathbf{1}_{\{u \mid mj+h\}} - \frac{1}{u} \quad (67)$$

and

$$K_{u,v}^{m_1, m_2}(j) = \mathbf{1}_{\{(j, H)=1\}} C_{m_1, u}(j) C_{m_2, v}(j). \quad (68)$$

This kernel has period $T = Hq$. Let

$$\widehat{K}(r) = \frac{1}{T} \sum_{x \bmod T} K_{u,v}^{m_1, m_2}(x) e_T(-rx). \quad (69)$$

Theorem 5.2 (Exact joint, single, and constant Fourier coefficients). *Let $\chi = \mathbf{1}_{\{g \mid m_1 - m_2\}}$. If $\chi = 1$, let $\kappa \bmod q$ be the class in [lemma 5.1](#); when $\chi = 0$, the first term below is omitted. For every $r \bmod Hq$,*

$$\begin{aligned} \widehat{K}(r) = \frac{c_H(r)}{Huv} & \left[g\chi e_q(-r\overline{H}\kappa) - \mathbf{1}_{\{b \mid r\}} e_u\left(-\frac{r}{b}\overline{H}\kappa_1\right) \right. \\ & \left. - \mathbf{1}_{\{a \mid r\}} e_v\left(-\frac{r}{a}\overline{H}\kappa_2\right) + \mathbf{1}_{\{q \mid r\}} \right]. \end{aligned} \quad (70)$$

In particular,

$$\widehat{K}(0) = \frac{\varphi(H)}{Huv}(g\chi - 1). \quad (71)$$

If $\chi = 1$, define

$$B_{g,a,b}(r) = g - \mathbf{1}_{\{b|r\}} - \mathbf{1}_{\{a|r\}} + \mathbf{1}_{\{q|r\}}. \quad (72)$$

Then the four terms have a common phase and

$$\boxed{\widehat{K}(r) = \frac{c_H(r)}{Huv} e_q(-r\overline{H}\kappa) B_{g,a,b}(r).} \quad (73)$$

Proof. Identify a class modulo Hq with $c \bmod H$ and $y \bmod q$. The additive character factors as

$$e_{Hq}(-rx) = e_H(-rc\overline{q})e_q(-ry\overline{H}).$$

Summation over $(c, H) = 1$ gives $c_H(r\overline{q}) = c_H(r)$, because q is a unit modulo H . The compatible joint class therefore contributes

$$\chi \frac{c_H(r)}{Hq} e_q(-r\overline{H}\kappa) = \frac{c_H(r)}{Huv} g\chi e_q(-r\overline{H}\kappa).$$

The class $y \equiv \kappa_1 \pmod{u}$ has $b = q/u$ lifts modulo q . Their character sum is zero unless $b \mid r$; when $r = bs$, it is $be_u(-s\overline{H}\kappa_1)$. After multiplying by the centering coefficient $-1/v$, this is the second term of (70). The other single term is identical with u, v interchanged. The constant term is supported on $q \mid r$. This proves the first formula and its zero mode.

Suppose now that $\chi = 1$. If $b \mid r$, reduction of the joint phase modulo u gives

$$e_q(-r\overline{H}\kappa) = e_u\left(-\frac{r}{b}\overline{H}\kappa_1\right).$$

The analogous identity holds when $a \mid r$, and the joint phase is one when $q \mid r$. Factoring it out gives (73). $\square$

Remark 5.3 (Sign convention). The coefficient $\widehat{K}(r)$ uses $e_{Hq}(-rx)$, hence the minus sign in (70). Direct Poisson summation of a progression uses $e_{Hq}(rx)$, and therefore has the conjugate phase. The two conventions are equivalent but should not be mixed in one formula.

The next form is convenient because it retains the natural period of each of the four channels.

Lemma 5.4 (Primitive progression Poisson formula). *Let $(s, H) = 1$, let $\rho \bmod s$, and let $F \in C_c^\infty(\mathbb{R})$. With $\widehat{F}(\xi) = \int_{\mathbb{R}} F(x)e(-x\xi) dx$,*

$$\sum_{\substack{j \in \mathbb{Z} \\ (j, H) = 1 \\ j \equiv \rho \pmod{s}}} F(j/J) = \frac{J}{Hs} \sum_{n \in \mathbb{Z}} c_H(n) e_s(n\overline{H}\rho) \widehat{F}\left(\frac{nJ}{Hs}\right). \quad (74)$$

Proof. Combine each reduced class modulo H with $\rho \bmod s$, apply Poisson summation to the resulting progressions modulo Hs , and sum the phases over the reduced classes modulo H . $\square$

Corollary 5.5 (Four natural lattices and one common lattice). *Under the hypotheses of theorem 5.2,*

$$\begin{aligned} & \sum_{j \in \mathbb{Z}} K_{u,v}^{m_1, m_2}(j) F(j/J) \\ &= \frac{J}{Huv} \left[g\chi \sum_{n \in \mathbb{Z}} c_H(n) e_q(n\overline{H}\kappa) \widehat{F}\left(\frac{nJ}{Hq}\right) \right. \end{aligned}$$

$$\begin{aligned}
& - \sum_{n \in \mathbb{Z}} c_H(n) e_u(n \overline{H} \kappa_1) \widehat{F}\left(\frac{nJ}{Hu}\right) \\
& - \sum_{n \in \mathbb{Z}} c_H(n) e_v(n \overline{H} \kappa_2) \widehat{F}\left(\frac{nJ}{Hv}\right) + \sum_{n \in \mathbb{Z}} c_H(n) \widehat{F}\left(\frac{nJ}{H}\right) \Big], \tag{75}
\end{aligned}$$

where the joint sum is omitted when $\chi = 0$. If $\chi = 1$, this equals

$$\boxed{\frac{J}{Huv} \sum_{r \in \mathbb{Z}} c_H(r) e_q(r \overline{H} \kappa) B_{g,a,b}(r) \widehat{F}\left(\frac{rJ}{Hq}\right)}. \tag{76}$$

Proof. Expand the product of the two centered indicators and apply [lemma 5.4](#) with moduli $q, u, v, 1$. In the compatible case, lift the u -single frequency n to $r = bn$, the v -single frequency to $r = an$, and the constant frequency to $r = qn$. Since $(abq, H) = 1$, multiplication by any of these lifting factors leaves c_H unchanged. The phase alignments in the proof of [theorem 5.2](#) then give (76). $\square$

Proposition 5.6 (Exact multiplier and kernel energy). *In the compatible case,*

$$0 \leq B_{g,a,b}(r) \leq g \tag{77}$$

and

$$\sum_{r \bmod q} B_{g,a,b}(r)^2 = qg^2 + (1 - 2g)(u + v) + 4g - 3. \tag{78}$$

Exactly

$$q - u - v + g = g(a - 1)(b - 1) \tag{79}$$

frequencies satisfy $B_{g,a,b}(r) = g$. More generally, with $\chi = \mathbf{1}_{\{g \mid m_1 - m_2\}}$,

$$\begin{aligned}
\sum_{r \bmod Hq} |\widehat{K}(r)|^2 &= \frac{\varphi(H)}{H} \left[\frac{(u-1)(v-1)}{u^2 v^2} \right. \\
&\quad \left. + \frac{g\chi - 1}{uv} \left(1 - \frac{2}{u}\right) \left(1 - \frac{2}{v}\right) \right]. \tag{80}
\end{aligned}$$

Proof. For $g = 1$, coprimality of a, b gives

$$B_{1,a,b}(r) = (1 - \mathbf{1}_{\{a \mid r\}})(1 - \mathbf{1}_{\{b \mid r\}}).$$

For $g \geq 2$, the possible values are $g, g-1, g-2$, proving (77). The counts of $b \mid r$, $a \mid r$, their intersection, and $q \mid r$ modulo q are respectively $u, v, g, 1$. Expanding the square proves (78); inclusion-exclusion proves (79).

For the last identity use Plancherel on $\mathbb{Z}/(Hq)\mathbb{Z}$. Conditional on $(j, H) = 1$, the two divisibility events have probabilities $1/u, 1/v$ and joint probability $g\chi/(uv)$. Writing a Bernoulli square as $(I - p)^2 = (1 - 2p)I + p^2$ gives the bracket in (80); the primitive mask contributes the density $\varphi(H)/H$. $\square$

6 Single-leg completion and the intermediate exponent wedge

The four natural lattices in [equation \(75\)](#) have different periods. This allows both single-indicator channels to be completed even when the joint period $H[u, v]$ is long.

Theorem 6.1 (Two-scale completion). *Fix $\eta > 0$, and let F range over a uniformly bounded subset of $C_c^\infty(\mathbb{R})$. Suppose*

$$H \max(u, v) \leq JX^{-\eta}. \quad (81)$$

Then, for every $A > 0$,

$$\begin{aligned} \sum_{j \in \mathbb{Z}} K_{u,v}^{m_1, m_2}(j) F(j/J) &= J \hat{F}(0) \frac{\varphi(H)}{Huv} (g\chi - 1) \\ &+ \chi \frac{J}{Hq} \sum_{n \neq 0} c_H(n) e_q(n\overline{H}\kappa) \hat{F}\left(\frac{nJ}{Hq}\right) + O_{A, \eta, F}(X^{-A}), \end{aligned} \quad (82)$$

where $q = [u, v]$, $\chi = \mathbf{1}_{\{g|m_1 - m_2\}}$, and the second line is absent when $\chi = 0$. The error remains arbitrarily power-saving after any fixed polynomial number of row and finite-model divisor sums carrying polynomially bounded weights.

Proof. In (75), every nonzero frequency of the u -single channel has Fourier argument of size at least $J/(Hu) \geq X^\eta$, and similarly for the v -single channel. The constant channel has the still shorter period H . Repeated integration by parts removes all their nonzero frequencies with arbitrary power saving. Their three zero modes add to $-J\hat{F}(0)\varphi(H)/(Huv)$. The joint zero mode is $J\hat{F}(0)\varphi(H)g\chi/(Huv)$, and its nonzero frequencies are retained. Choosing the number of integrations after the fixed polynomial multiplicity proves the last assertion. $\square$

Corollary 6.2 (Low-frequency joint packet). *Under the hypotheses of [theorem 6.1](#), fix $0 < \epsilon < \eta$. The joint remainder in (82) may be truncated to*

$$0 < |n| \leq N_{q, \epsilon}, \quad N_{q, \epsilon} = X^\epsilon \frac{Hq}{J}, \quad (83)$$

with arbitrary power error. Whenever this range is nonempty,

$$|n| < a, \quad |n| < b, \quad (84)$$

for all sufficiently large X . Consequently the compatible common-lattice multiplier satisfies

$$B_{g, a, b}(n) = g \quad (0 < |n| \leq N_{q, \epsilon}). \quad (85)$$

Proof. For $|n| > N_{q, \epsilon}$, the Fourier argument in the joint sum has size greater than X^ϵ , so rapid decay of $\hat{F}$ gives the truncation. Since $q/b = u$ and $q/a = v$,

$$\frac{N_{q, \epsilon}}{b} = X^\epsilon \frac{Hu}{J} \leq X^{\epsilon - \eta}, \quad \frac{N_{q, \epsilon}}{a} = X^\epsilon \frac{Hv}{J} \leq X^{\epsilon - \eta}.$$

Thus a nonzero retained integer is divisible by neither a nor b . It is not divisible by q , and (72) gives (85). $\square$

Remark 6.3 (What matching does in this wedge). In the regime of [corollary 6.2](#), matching the four channels completes their common zero mode, but it produces no further algebraic cancellation on the retained nonzero frequencies: the multiplier there is exactly g . The remaining joint packet therefore requires cancellation in the row variables, the divisor variables, or both. This is a low-frequency no-go statement, not a lower bound for the full TPC-18 correlation.

We now translate the new condition into the TPC scale. Recall

$$R = X^{1/2 - \delta + o(1)}, \quad J \asymp X/Q, \quad 0 < \delta < \frac{1}{2}. \quad (86)$$

Corollary 6.4 (Whole finite-model single-leg range). *Fix $\eta_0 > 0$. If*

$$Q \leq X^{1/2+\delta-\eta_0}, \quad (87)$$

then, for every fixed $0 < \eta < \eta_0$ and all sufficiently large X , condition (81) holds simultaneously for every $u, v \leq R$. Thus every finite-model single and constant nonzero spectrum is power-saving, and only compatible joint packets can remain long.

By comparison, completing every joint period by smoothness alone requires

$$Q \leq X^{2\delta-\eta_0}. \quad (88)$$

Hence the intermediate wedge

$$X^{2\delta+\eta_1} < Q \leq X^{1/2+\delta-\eta_0} \quad (89)$$

is nonempty for fixed sufficiently small positive margins η_0, η_1 .

Proof. Uniformly for $u, v \leq R$,

$$\frac{H \max(u, v)}{J} \ll_h Q X^{-1/2-\delta+o(1)}.$$

This is at most $X^{-\eta}$ under (87). On the other hand, $H[u, v]/J \leq HR^2/J \ll_h Q X^{-2\delta+o(1)}$, which gives (88). Since $1/2 + \delta - 2\delta = 1/2 - \delta > 0$, fixed margins leave a nonempty wedge. $\square$

Corollary 6.5 (Endpoint ledger for the largest divisor slice). *Write $L = X^{\lambda+o(1)}$ and $D \leq V = X^{1/4-\delta/2+o(1)}$. The largest row scale satisfies*

$$Q \leq X^{\lambda+1/4-\delta/2+o(1)}. \quad (90)$$

It lies strictly inside the single-leg range whenever

$$\delta > \frac{2}{3} \left(\lambda - \frac{1}{4} \right) \quad (91)$$

with a fixed margin. Thus the published profile inherited from TPC-17 and TPC-18 [8, 9], $\lambda = 10/21 - \sigma$, has threshold

$$\delta > \frac{19}{126} - \frac{2}{3}\sigma, \quad (92)$$

and the version-locked Li-v6 profile [6, 8, 9], $\lambda = 8/17 - \sigma$, has threshold

$$\delta > \frac{5}{34} - \frac{2}{3}\sigma. \quad (93)$$

All three displayed inequalities are strict boundary statements; equality does not supply the power margin used by theorem 6.1.

Proof. Compare the exponent in (90) with $1/2 + \delta$. The resulting inequality is $\lambda - 1/4 < 3\delta/2$. Substitution of the two values of λ gives (92) and (93). $\square$

Proposition 6.6 (Sharp boundary of smooth single-leg completion). *Polynomial separation $Hu \leq JX^{-\eta}$ is the sufficient condition used above. It cannot be replaced by the comparable-scale regime using rapid Fourier decay alone. More precisely, for fixed $0 < c < C < \infty$, when $c \leq J/(Hu) \leq C$, there exist a fixed smooth compactly supported F and an allowed class $\kappa_1 \bmod u$ for which the first nonzero term of the unnormalized primitive progression*

$$\frac{J}{Hu} c_H(1) e_u(\overline{H}\kappa_1) \hat{F}\left(\frac{J}{Hu}\right) \quad (94)$$

has size bounded below by a positive constant depending on H, F . Consequently no extension beyond the exponent boundary $Q = X^{1/2+\delta+o(1)}$ follows from smoothness alone.

Proof. Choose a nonnegative, nonzero smooth F , supported sufficiently close to zero that

$$\Re \widehat{F}(\xi) \geq \frac{1}{2} \int_{\mathbb{R}} F(t) dt > 0 \quad (c \leq \xi \leq C).$$

Since H is squarefree, $c_H(1) = \mu(H) \neq 0$, and the phase has modulus one. The displayed mode is therefore bounded below uniformly in the stated range and is not power-saving. This is a boundary for the termwise completion method; cancellation against other arithmetic sums would be an additional input. $\square$

Remark 6.7 (Scope). The kernel theorem is coefficient-independent. Applied with λ'_R , it concerns the centered finite model; applied with $b_R \mathbf{1}_{u \leq R}$, it concerns the short–short block of the matched residual. It neither estimates the remaining compatible joint packet nor controls the residual terms with $u > R$ or $v > R$, and therefore does not prove the full primitive residual-dispersion target.

7 CRT-fiber compression and the exact remaining gate

We now compress the compatible joint inverse phases. Throughout this section the row and orbit weights have first been separated as in the smooth-separation step of the TPC interface, so that the same function $F(j/J)$ occurs for every row pair in one packet. The separated row weights are absorbed into coefficients z_{m_1, m_2} . The argument is exact for each separated packet; the usual rapidly convergent superposition may be restored afterwards.

Fix u, v with $(uv, H) = 1$, put $q = [u, v]$, and let $\mathcal{R}_{u,v}$ be any finite set of row pairs such that $(u, m_1) = (v, m_2) = 1$ for every $(m_1, m_2) \in \mathcal{R}_{u,v}$, including any generic pruning independent of j . Retain only the compatible pairs $g \mid m_1 - m_2$.

Lemma 7.1 (Compatible row residues are CRT fibers). *The map which assigns to a compatible row pair its joint class $\kappa \bmod q$ takes values in*

$$\mathcal{U}_q = (\mathbb{Z}/q\mathbb{Z})^\times. \quad (95)$$

Conversely, a class $\kappa \in \mathcal{U}_q$ determines the unique compatible pair of row residue classes

$$m_1 \equiv -h\bar{\kappa} \pmod{u}, \quad m_2 \equiv -h\bar{\kappa} \pmod{v}. \quad (96)$$

Thus compatible pairs of row residue classes are in bijection with $\mathcal{U}_q$.

Proof. The forward statement is [lemma 5.1](#). Conversely, (96) makes $m_i \kappa \equiv -h$ in the required modulus. The two resulting classes are units and are compatible by construction. Inversion modulo u and v shows uniqueness. $\square$

Define the fiber coefficient

$$Z_{u,v}(\kappa) = \sum_{\substack{(m_1, m_2) \in \mathcal{R}_{u,v} \\ m_1 \equiv -h\bar{\kappa} \pmod{u} \\ m_2 \equiv -h\bar{\kappa} \pmod{v}}} z_{m_1, m_2}, \quad \kappa \in \mathcal{U}_q, \quad (97)$$

and its joint phase packet

$$S_{u,v}(r) = \sum_{\kappa \in \mathcal{U}_q} Z_{u,v}(\kappa) e_q(r \bar{H} \kappa). \quad (98)$$

Theorem 7.2 (Triple inverse phases compress to one DFT). *The sum over compatible row pairs of the joint Poisson phase is exactly $S_{u,v}(r)$. If u and v are squarefree, as they are on both the λ'_R - and b_R -supports, then in the coordinates $u = ga$, $v = gb$ the integers g, a, b are pairwise coprime and each summand may equivalently be written as*

$$e_q(r\overline{H}\kappa) = e_g\left(-hr\overline{H}\overline{m_1}ab\right) e_a\left(-hr\overline{H}\overline{m_1}gb\right) \times e_b\left(-hr\overline{H}\overline{m_2}ga\right). \quad (99)$$

with a modulus-one factor interpreted as 1. Thus the three displayed inverse phases are the CRT factorization of one ordinary Fourier character, not three independent spectra. Moreover,

$$\boxed{\sum_{r \bmod q} |S_{u,v}(r)|^2 = q \sum_{\kappa \in \mathcal{U}_q} |Z_{u,v}(\kappa)|^2.} \quad (100)$$

Proof. The first assertion follows by grouping the row pairs according to the bijection in [lemma 7.1](#). Under the stated squarefree hypothesis, the factorization in (99) is CRT factorization of the character modulo the pairwise coprime moduli g, a, b , followed by $m_i\kappa \equiv -h$ in the appropriate factors. Finally, $r \mapsto r\overline{H}$ permutes the frequencies modulo q , so ordinary finite-group Parseval proves (100). $\square$

All pairs (u, v) with the same least common multiple may be combined before estimation. In the single-leg range of [corollary 6.4](#), let $\vartheta_R(u)$ denote either the finite-model coefficient $\lambda'_R(u)$, or the matched short-short residual coefficient $b_R(u)\mathbf{1}_{u \leq R}$, and put

$$Z_q(\kappa) = \sum_{\substack{u, v \leq R \\ \text{lcm}(u, v) = q}} \vartheta_R(u)\vartheta_R(v)Z_{u,v}(\kappa), \quad (101)$$

where an inadmissible row-modulus pair contributes zero. Define

$$S_q(r) = \sum_{\kappa \in \mathcal{U}_q} Z_q(\kappa)e_q(r\overline{H}\kappa). \quad (102)$$

Then the compatible long-joint contribution at this denominator is

$$\mathcal{L}_q = \frac{J}{Hq} \sum_{r \neq 0} c_H(r) \widehat{F}\left(\frac{rJ}{Hq}\right) S_q(r). \quad (103)$$

This formula is a normal form for the chosen short-short joint remainder. In the residual application it does not include the large-divisor legs $u > R$ or $v > R$.

Theorem 7.3 (Ramanujan mean and fiber discrepancy). *Put*

$$\overline{Z}_q = \frac{1}{\varphi(q)} \sum_{\kappa \in \mathcal{U}_q} Z_q(\kappa), \quad D_q(\kappa) = Z_q(\kappa) - \overline{Z}_q. \quad (104)$$

Then

$$S_q(r) = \overline{Z}_q c_q(r) + \sum_{\kappa \in \mathcal{U}_q} D_q(\kappa) e_q(r\overline{H}\kappa), \quad (105)$$

and

$$\boxed{\sum_{r \bmod q} |S_q(r) - \overline{Z}_q c_q(r)|^2 = q \sum_{\kappa \in \mathcal{U}_q} |D_q(\kappa)|^2.} \quad (106)$$

Since $(H, q) = 1$, the deterministic part of (103) is the one-modulus Ramanujan packet

$$\mathcal{L}_q^{\text{mean}} = \overline{Z}_q \frac{J}{Hq} \sum_{r \neq 0} c_{Hq}(r) \widehat{F}\left(\frac{rJ}{Hq}\right). \quad (107)$$

Proof. Insert $Z_q = \overline{Z}_q + D_q$ in (102). The Fourier transform of the constant function on $\mathcal{U}_q$ is $c_q(r\overline{H}) = c_q(r)$. Parseval applied to D_q , extended by zero off $\mathcal{U}_q$, gives (106). Multiplicativity of Ramanujan sums for coprime moduli gives $c_H(r)c_q(r) = c_{Hq}(r)$. $\square$

Proposition 7.4 (Restricted-frequency fiber bound). *Fix $\epsilon > 0$ and suppose*

$$J \geq 4HX^\epsilon. \quad (108)$$

Write $\mathcal{L}_{q, \leq N}^{\text{disc}}$ for the discrepancy part of (103) truncated at

$$0 < |r| \leq N_{q, \epsilon} = X^\epsilon \frac{Hq}{J}, \quad (109)$$

Then

$$\left| \mathcal{L}_{q, \leq N}^{\text{disc}} \right| \ll_{H, F, \epsilon} X^{\epsilon/2} \sqrt{\frac{J}{H}} \left(\sum_{\kappa \in \mathcal{U}_q} |D_q(\kappa)|^2 \right)^{1/2} \quad (110)$$

and, for every $A > 0$,

$$\mathcal{L}_q^{\text{disc}} = \mathcal{L}_{q, \leq N}^{\text{disc}} + O_{A, F, H, \epsilon} \left(X^{-A} \sum_{\kappa \in \mathcal{U}_q} |D_q(\kappa)| \right). \quad (111)$$

Let $\mathcal{Q}$ be a family of long denominators and assume, for some fixed C_0 , the explicit polynomial mass bound

$$\sum_{q \in \mathcal{Q}} \sum_{\kappa \in \mathcal{U}_q} |D_q(\kappa)| \ll X^{C_0}. \quad (112)$$

Then a sufficient discrepancy input is

$$\sum_{q \in \mathcal{Q}} \left(\sum_{\kappa \in \mathcal{U}_q} |D_q(\kappa)|^2 \right)^{1/2} \ll X^{-\epsilon} \frac{XQ}{\sqrt{J}} (\log X)^{-B}, \quad (113)$$

with the harmless power of X^ϵ adjusted as needed. This controls only the discrepancy packets; the Ramanujan mean packets in (107) remain separate.

Proof. Rapid decay of $\widehat{F}$, the trivial bound

$$\left| \sum_{\kappa \in \mathcal{U}_q} D_q(\kappa) e_q(r\overline{H}\kappa) \right| \leq \sum_{\kappa \in \mathcal{U}_q} |D_q(\kappa)|,$$

and summation of the Fourier tail give (111). Under (108), the retained positive and negative integers occupy distinct classes modulo q . Cauchy-Schwarz, $|c_H(r)| \ll_H 1$, and (106) give

$$\begin{aligned} |\mathcal{L}_{q, \leq N}^{\text{disc}}| &\ll_H \frac{J}{Hq} N_{q, \epsilon}^{1/2} \left(q \sum_{\kappa \in \mathcal{U}_q} |D_q(\kappa)|^2 \right)^{1/2} \\ &\ll_H X^{\epsilon/2} \sqrt{\frac{J}{H}} \left(\sum_{\kappa \in \mathcal{U}_q} |D_q(\kappa)|^2 \right)^{1/2}. \end{aligned}$$

Summation over q proves the sufficient gate after choosing the epsilon bookkeeping with the required margin. Equation (112) makes the summed tails negligible by taking $A > C_0 + B + 1$ in (111). $\square$

The Ramanujan mean has a revealing interaction with the transverse Möbius sign. The following identity applies to each fixed squarefree (u, v) -summand before the lcm aggregation in (101); it is not an estimate.

Proposition 7.5 (Migration of the transverse sign). *Assume $q = gab$ is squarefree with g, a, b pairwise coprime. Put*

$$d_g = (r, g), \quad d_a = (r, a), \quad d_b = (r, b). \quad (114)$$

Then

$$\boxed{\mu(a)\mu(b)c_{gab}(r) = \mu(g/d_g)\mu(d_a)\mu(d_b)\varphi(d_g d_a d_b).} \quad (115)$$

In particular,

$$(r, gab) = 1 \implies \mu(a)\mu(b)c_{gab}(r) = \mu(g). \quad (116)$$

Thus fiber-uniformity of a fixed squarefree (u, v) -summand does not preserve its transverse Möbius sign; on frequencies coprime to q , that sign is transferred to the associated determinant content $g = (u, v)$. No single- g conclusion is asserted after summing over all decompositions of q .

Proof. For squarefree q ,

$$c_q(r) = \mu\left(\frac{q}{(q, r)}\right) \varphi((q, r)).$$

Factor this formula over the pairwise coprime integers g, a, b . Since $\mu(a)\mu(a/d_a) = \mu(d_a)$, and similarly for b , one obtains (115). $\square$

Proposition 7.6 (Sharp algebraic no-go for arbitrary fibers). *CRT compression alone implies neither uniform pointwise decay of the restricted discrepancy transform nor a restricted-frequency saving for arbitrary fiber vectors. Indeed, if an admissible coefficient vector is supported on one class $\kappa_0 \in \mathcal{U}_q$, say $Z_q = A\mathbf{1}_{\{\kappa_0\}}$, then*

$$|S_q(r)| = |Z_q(\kappa_0)| \quad \text{for every } r. \quad (117)$$

Moreover its mean-zero discrepancy transform is

$$\widehat{D}_q(r) = A \left(e_q(r\overline{H}\kappa_0) - \frac{c_q(r)}{\varphi(q)} \right). \quad (118)$$

If $\varphi(q) > 1$ and $(r, q) = 1$, then

$$|\widehat{D}_q(r)| \geq |A| \left(1 - \frac{1}{\varphi(q)} \right). \quad (119)$$

Thus a nonempty retained window, which contains the coprime frequency $r = 1$, can still carry order- $|A|$ discrepancy. Its uncentered Fourier energy is exactly $q|A|^2$, distributed uniformly over all q frequencies. A useful improvement of (110) requires an arithmetic bound for the actual row-fiber discrepancy, or a separable low-rank decomposition whose coefficient cost is controlled.

Proof. For a one-fiber vector, $S_q(r) = Z_q(\kappa_0)e_q(r\overline{H}\kappa_0)$. This proves (117) and equality in the corresponding Parseval identity. Subtracting the unit-class mean gives (118). When $(r, q) = 1$, $c_q(r) = \mu(q)$, so the reverse triangle inequality proves (119). $\square$

Remark 7.7 (Why a generic additive large sieve does not close the gate). For denominators $q \asymp Q_0$, the relevant dual length is $N \asymp HQ_0/J$, whereas generic Farey spacing costs Q_0^2 . Thus the standard large-sieve constant $N + Q_0^2$ is dominated by Q_0^2 in the present low-frequency regime. Likewise, existing bilinear or trilinear Kloosterman bounds require a separated coefficient array and do not accept an arbitrary modulus-dependent vector $Z_q(\kappa)$. The exact new gate is therefore a quantitative equidistribution or controlled-rank theorem for the actual prime-Möbius row fibers, together with control or matching of the explicit Ramanujan mean packet.

Remark 7.8 (Scope of the compression). The compression theorem applies to a separated compatible joint packet. In the intermediate range of [corollary 6.4](#), it describes the only nonzero spectrum left in the chosen short-short block after single-leg completion. It does not by itself estimate that packet, control the large-divisor legs of the common residual coefficient, or prove the all-modulus primitive residual-dispersion target.

8 Technology interfaces and the coefficient gate

TPC-18 produces the primitive two-row residual correlation, and TPC-19 separates its direct determinant-frequency and long-lcm Poisson interfaces [\[9, 10\]](#). The results above improve the second interface in two exact ways: the four centered channels are placed on one Fourier lattice before estimation, and the compatible triple inverse phase is compressed to the single fiber transform [\(102\)](#). This section records how several cited estimates interface with those reductions, and which hypothesis on the chosen prime-Möbius fiber vector is still missing.

8.1 The frequency-concentration interface

A particularly flexible input relevant here is the exceptional-spectrum large sieve of Pascadi [\[7\]](#). Put $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. In its notation, for $\alpha, \beta \in \mathbb{T}$, put

$$T_N(\alpha, \beta) = \min_{t \in \mathbb{Z}_{\geq 1}} (t + N\|t\alpha\|_{\mathbb{T}} + N\|t\beta\|_{\mathbb{T}}), \quad \mathcal{I}_N(\nu) = \iint T_N(\alpha, \beta) d|\nu|(\alpha) d|\nu|(\beta). \quad (120)$$

Theorem 5.2 of that paper treats a coefficient $a_n = f(n/N)\check{\nu}(n)$. Suppressing its standard smooth-cutoff, parameter, and cusp-normalization hypotheses, the theorem gives

$$\sum_{\lambda_j < 1/4} X^{2\theta_j} \left| \sum_{n \lesssim N} a_n \rho_{ja}(a_0 n) \right|^2 \ll_{\epsilon} (qa_0 N X)^{2\epsilon} \left(1 + \frac{a_0 N}{q} \right) \mathcal{A}^2, \quad (121)$$

provided, in particular,

$$\mathcal{A} \gg \|a\|_2 + \frac{\sqrt{(a_0, q)}N}{\sqrt{q + a_0 N}} |\nu|(\mathbb{T}), \quad (122)$$

$$X \ll \max\left(1, \frac{q}{a_0 N}\right) \max\left(1, \frac{\mathcal{A}^2}{\mathcal{I}_N(\nu)}\right). \quad (123)$$

Thus this result is an interface theorem, not a black-box estimate for an arbitrary determinant sequence. Before it can be used here one must derive one common spectral coefficient from the matched four-channel kernel, identify its spectral level q , length N , scalar a_0 , and exceptional parameter X , construct a representing measure ν , and verify both [\(122\)](#) and [\(123\)](#). In particular, the spectral level called q in [\(121\)](#) must not be identified with either the row scale or the fiber modulus merely because the same letter is traditionally used. Applying the theorem separately to the joint, two single, and constant channels is not a substitute: the intervening triangle inequalities would discard the matching encoded by [\(73\)](#). The coefficient presented to the large sieve must be the combined coefficient whenever more than one channel survives on the block under consideration.

There is a sharp positive benchmark. Theorem 1.7 of Pascadi [\[7\]](#) applies to the smooth difference coefficient

$$\sum_{h_1 \ell_1 - h_2 \ell_2 = n} \Phi_1(h_1/H_0) \Phi_2(h_2/H_0) e(h_1 \alpha_1 + h_2 \alpha_2),$$

subject to its stated spectral-level and parameter conditions, including the regime in which the level is at least of order L^2 . It does not cover the sharp coefficient isolated in TPC-19, where ω_D is a 1-bounded smooth cutoff supported on $d \asymp D$:

$$A_{\ell_1, \ell_2}(n) = \sum_{\ell_1 d_1 - \ell_2 d_2 = n} \mu(d_1) \mu(d_2) \omega_D(d_1) \omega_D(d_2). \quad (124)$$

Indeed, if

$$M_\ell(\alpha) = \sum_d \mu(d) \omega_D(d) e(-\ell d \alpha),$$

then $\widehat{A}_{\ell_1, \ell_2} = M_{\ell_1} \overline{M_{\ell_2}}$. The natural representing measure has density $\widehat{A}_{\ell_1, \ell_2}$, and hence total variation

$$V_{\ell_1, \ell_2} = \int_{\mathbb{T}} |M_{\ell_1}(\alpha) M_{\ell_2}(\alpha)| d\alpha \ll D \quad (125)$$

by Parseval and Cauchy–Schwarz. Indeed, $\|M_\ell\|_2^2 \leq \sum_{d \asymp D} |\omega_D(d)|^2 \ll D$. In the formal smooth-benchmark scaling $N \asymp LD$, spectral level $q \asymp L^2$, $a_0 = 1$, and $L > 2D$, retaining $\mathcal{A} = D^{1+o(1)}$ would already require $V_{\ell_1, \ell_2} = D^{o(1)}$, together with the independent two-frequency condition (123). Neither estimate follows from (125).

Davenport’s uniform Möbius exponential-sum estimate, in the form recalled by Green and Tao [4], gives arbitrary logarithmic saving in $\sup_\alpha |\sum_{d \leq D} \mu(d) e(d\alpha)|$. After partial summation it also handles a standard smooth dyadic cutoff. This is a one-frequency L^∞ statement. It does not bound the product variation in (125) at the scale required above and does not control the two-frequency functional $\mathcal{I}_N$. Consequently it does not verify Pascadi’s two gates for (124).

8.2 What one-fraction estimates do and do not import

The classical bilinear estimate of Duke et al. [3] controls a single Kloosterman fraction $e(am/n)$ with a separated coefficient $\alpha_m \beta_n$. The trilinear estimate of Bettin and Chandee [1] admits a further separated coefficient ν_a , and the recent preprint Wright [11] treats a useful partially fixed denominator while retaining the same essential one-fraction and separability requirements. The latter is presently a version-sensitive, non-peer-reviewed preprint, so only its explicitly stated current hypotheses may be used. These results can therefore be invoked on a dyadic slice only after the slice has been reduced to their precise phase and coefficient forms and all of their size hypotheses have been checked. They are not estimates for an arbitrary two-dimensional array indexed by a modulus and a residue class.

The compression in theorem 7.2 supplies the needed single phase, but not the needed coefficient separation. After aggregation, the relevant vector is

$$(q, \kappa) \longmapsto D_q(\kappa), \quad \kappa \in \mathcal{U}_q.$$

Replacing κ by its inverse may display the phase as one Kloosterman fraction, but it simultaneously replaces the coefficient by $D_q(\overline{m})$; the joint dependence on q and m remains. Likewise, Theorem 1 of Blomer and Milićević [2] averages complete Kloosterman sums over their modulus c , with a residue weight modulo q_0 , a smooth c/X weight, and a cost measured by the ℓ^1 -norm of the q_0 -Mellin transform of that residue weight. It is not a direct estimate for the one-modulus fiber transform $S_q(r)$. The identity (102) alone does not convert the fiber vector into that weighted-modulus average. None of the cited results is a separate *long-lcm Poisson spectrum* theorem: Poisson summation gives the exact normal form, while the arithmetic estimate of its coefficient array is an additional step.

The obstruction may be quantified. On a dyadic modulus block $q \asymp Q_0$, choose representatives $0 \leq \kappa < 2Q_0$, extend $D_q(\kappa)$ by zero outside $\kappa \in \mathcal{U}_q$, and define the separable projective cost

$$\mathfrak{R}_\pi(D; Q_0) = \inf_{D_q(\kappa) = \sum_\nu x_\nu(q) y_\nu(\kappa)} \sum_\nu \|x_\nu\|_2 \|y_\nu\|_2. \quad (126)$$

A direct termwise use of a separated bilinear or trilinear theorem pays the decomposition cost in (126). Small Hilbert-space norm or Parseval energy does not by itself imply a sufficiently small projective cost. Thus “one phase” is a necessary structural reduction, whereas controlled projective rank is an additional arithmetic input.

8.3 A defensible next gate

For a family $\mathcal{Q}$ of long denominators, set

$$\mathfrak{V}(\mathcal{Q}) = \sum_{q \in \mathcal{Q}} q \sum_{\kappa \in \mathcal{U}_q} |D_q(\kappa)|^2. \quad (127)$$

By (106), this is exactly the complete frequency energy of the discrepancy transforms. It is not yet a restricted-frequency saving: the one-fiber example in proposition 7.6 retains an order-one coprime-frequency discrepancy after its mean is removed. The direct next proof obligation is therefore the weighted $\ell_q^1 \ell_\kappa^2$ estimate stated in (113), or a stronger variance estimate that implies it with a fixed margin. This route uses the exact restricted-frequency bound in proposition 7.4 and requires no claim that any cited Kloosterman theorem accepts the fiber array.

An alternative low-rank route must do all three of the following on each dyadic block: exhibit an explicit decomposition in (126); substitute the actual block parameters into one specified theorem of Duke et al. [3], Bettin and Chandee [1], or Wright [11]; and prove that the imported bound multiplied by the full decomposition cost is below the TPC-18 obstruction scale with a fixed margin. Low rank without this norm and parameter ledger is not a valid transfer theorem. In either route the explicit Ramanujan mean packet (107) remains a separate term; fiber discrepancy estimates do not cancel it.

Finally, no estimate in this paper is taken from the January 2026 withdrawn preprint entitled *Bilinear forms with Kloosterman fractions and applications*. Its withdrawal notice reports a missing factor in the claimed bound, and it is intentionally given no bibliographic entry here. The unconditional boundary is therefore as follows: the results audited above supply rigorous smooth and separated benchmarks, while neither the sharp frequency-concentration gates nor the actual fiber-variance/projective-rank gate is proved here or supplied by those results. In particular, the results above do not establish cancellation of the full TPC-18 residual, a fixed-shift prime-pair asymptotic, a twin-prime lower bound, or a breach of the sieve parity barrier.

9 Finite dynamics, exact certificate, and conclusion

The word *spectrum* in this paper has a finite and literal dynamical meaning. For each divisor pair, translation

$$\tau : j \longmapsto j + 1 \quad \text{on } \mathbb{Z}/(H[u, v])\mathbb{Z} \quad (128)$$

is a finite measure-preserving dynamical system. Its Koopman characters are $e_{H[u, v]}(rj)$, and the coefficients computed in theorem 5.2 are exactly the spectral coordinates of the centered local divisor observable. The Poisson step couples these finite cyclic spectra to the smooth orbit window, while theorem 7.2 reorganizes the compatible row weights on a single CRT fiber. This terminology does not assert a spectral realization of the zeros of the Riemann zeta function.

9.1 Exact computational certificate

The standard-library Python certificate is located at

`experiments/tpc20_certificate.py.`

Run it from the paper directory with

```
python experiments/tpc20_certificate.py
python -O experiments/tpc20_certificate.py
```

Both commands regenerate the deterministic file `experiments/tpc20_certificate.json`. The proof checks use integers, `fractions.Fraction`, formal prime-log polynomials, and finite prime fields containing roots of the exact required orders. They verify independently sampled instances of:

- (i) the pointwise four-channel recombination into the single coefficient b_R , including its scalar calibration term;
- (ii) the primitive complete-period zero mode and its termwise disappearance for every coprime divisor pair;
- (iii) the full four-term Fourier formula, the aligned multiplier $B_{g,a,b}$, its frequency counts and exact energy, and the kernel Plancherel identity;
- (iv) the CRT-fiber parametrization, finite Fourier transform, Parseval formula, and mean-discrepancy decomposition;
- (v) the squarefree sign migration, completion exponent ledger, and the convolution identity underlying the uniform calibration theorem.

The certificate is a regression test for finite algebra; it is not numerical evidence for the unproved fiber-discrepancy estimate. Its JSON therefore records

```
twin_prime_evidence = false,    residual_dispersion_evidence = false,
asymptotic_evidence = false.
```

Its diagnostic scope also excludes a parity-barrier breakthrough.

At the version archived with this paper, the generated JSON has SHA-256

```
0F994F32950AC68B8249D2F90E7074176106C122A2AE53268370F03584B7C12B,
```

and the script has SHA-256

```
55D9FA7DD4277C541ABE39C53231453F7F6A9107DEF40103F5227373B53E0174.
```

9.2 Conclusion

TPC-20 supplies the matched reduction that TPC-19 left open. The prime–prime, prime–model, model–prime, and model–model channels combine pointwise into one centered divisor kernel with one residual coefficient. A uniform Möbius theorem makes the finite calibration defect arbitrarily logarithmically small throughout polynomial row ranges, and [corollary 4.2](#) makes it negligible at packet scale under its explicit packet hypotheses, which hold for the separated packets in the single-leg range. The resulting primitive-conditional zero mode is connected, and the zero mode of every coprime divisor pair vanishes before any triangle inequality is applied.

The exact Fourier calculation then separates genuinely different period scales. Completing the two single legs at periods Hu and Hv , rather than artificially at $H[u, v]$, identifies uniform single-leg completion up to $Q \leq X^{1/2+\delta-\eta_0}$, beyond the older whole-joint-period completion range $Q \leq X^{2\delta-\eta_0}$. This removes two spurious long spectra. It does not cancel the surviving joint packet: on its low nonzero frequencies the aligned multiplier equals $g = (u, v)$.

Finally, the apparent triple inverse phase of TPC-19 is one ordinary DFT after compatible row pairs are grouped by their joint CRT class. The transform splits exactly into a Ramanujan mean packet and a fiber discrepancy, with Parseval identifying the full discrepancy energy. For this chosen compatible short–short packet, the remaining obligations are concrete: prove

the weighted restricted-frequency fiber gate (or an explicitly normed low-rank substitute) for the actual sharp Möbius row weights, and separately control the Ramanujan mean. The cited separated Kloosterman-fraction and frequency-concentration results do not automatically provide either input. The large-divisor legs remain outside this reduction.

Thus this paper advances the route by an exact common-channel compression, a wider unconditional single-leg completion range, and a sharply formulated coefficient gate. It does not estimate the full TPC-18 residual, prove a fixed-shift Hardy–Littlewood asymptotic or a twin-prime lower bound, breach the sieve parity barrier, or realize zeta zeros as dynamical eigenvalues.

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